

DEPARTMENT OF METALLURGICAL AND
MATERIALS ENGINEERING

IIT PATNA



PLACEMENT BROCHURE

Master of Technology(M. Tech) in Materials Science and Engineering (MSE) 2020-21

"A vital phenomenon can only be regarded as explained if it has been proven that it appears as the result of the material components of living organisms interacting according to the laws which those same components follow in their interactions outside of living systems."

— Adolf Eugen Fick

For us, an atom shall be a small, spherical, homogeneous body or an essentially indivisible, material point, whereas a molecule shall be a separate group of atoms in any number and of any nature.

— Marc Antoine Augustin Gaudin

From packaging materials, through fibers, foams and surface coatings, to continuous extrusions and large-scale moldings, plastics have transformed almost every aspect of life. Without them, much of modern medicine would be impossible and the consumer electronics and computer industries would disappear. Plastic sewage and water pipes alone have made an immeasurable contribution to public health worldwide.

— Norman C. Billingham



CONTENTS

FROM THE HOD'S DESK	03
THE DEPARTMENT	04
THE FACULTY MEMBERS	05
STATE OF ART LAB FACILITIES	06
OUR FUNDERS	10
ACHIEVEMENTS	11
CURRICULUM	12
BATCH PROFILE SUMMARY	13
CONTACT US	14

From the HOD's desk

"The Department of Metallurgical and Materials Engineering (MME) at IIT Patna is endowed with world class facilities under a single roof which allow students to learn vital skills and obtain hands-on experience of latest technologies used both in industries and academia."

"This Department has pioneered a unified approach in teaching and research, which has enabled us to evolve into an interdisciplinary field catering to diverse applications."



The Department of Metallurgical and Materials Engineering (MME), earlier known as Materials Science and Engineering at IIT Patna was established in 2013.

From its inception, it has shown its strong impact in the areas of research as well as industrial exposure. The field of study in the Department encompasses areas of both traditional and modern Metallurgy and other allied systems of materials (e.g., ceramics, polymer, composites etc).

Our UG programme in Metallurgical and Materials Engineering has already commenced in July 2019. The PG programme (i.e., M.Tech. in Materials Science and Engineering), started in 2014, is more interdisciplinary in nature.

Along with metals and alloys, the Department focuses heavily on a variety of Materials like ceramics, polymers, intermetallics, composites, electronic materials and biomaterials.

The Department has reinvented constantly to keep the curriculum in pace with the state-of-the-art technologies. This Department has pioneered a unified approach in teaching and research, which has enabled us to evolve into an interdisciplinary field catering to diverse applications.

MME Department at IIT Patna is endowed with world class facilities under single roof which allow students to learn vital skills and obtain hands-on experience of latest technologies used both in industries and academia. Department is actively engaged with various funding agency like SERB-DST, ISRO, NRB, BRNS, Manali Petrochemical Ltd., Denka (Japan), Tata Steel Limited, MRF Tyres, Carborundum Universal Limited etc.

Dr. Anup Kumar Keshri,
Head, Department of MME

VISION

The Department aims to train students with the fundamentals and expertise in design, synthesis, characterization, testing and production of various materials and also with the knowledge of new technologies in production and processing of new and advanced engineering materials.

The Department continues to strive for excellence and develop continuously through progress

The Department



The strong dependence of our society on metals and alloys makes metallurgy an important branch of modern engineering. Metallurgical and materials engineering involves design, innovate and improve the process; to transform minerals and metals to useful products we use every day in our life. It is a discipline that enables both the creation and application of materials in society. Also, Materials engineers develop materials for new applications, improve existing materials to enhance performance and evaluate ways in which different materials can be used together.

Department of Materials Science and Engineering was established in the year 2012 and was renamed as Department of Metallurgical and Materials Engineering in 2018. The Department focuses on the fundamental and engineering aspects of conventional metallurgy, materials and advanced materials.

The Department faculty specializes in the areas of physical and mechanical metallurgy, phase transformations, thermomechanical processing, electron microscopy, materials chemistry, nanomaterials, thin films and coatings, ceramic and metal matrix composites, tribology, thermal spraying, polymer science and technology,

fillers, composites, ultra-high temperature ceramics and phase field modeling. Apart from the regular teaching activities, the Department aims to train students with the fundamentals and expertise of design, synthesis, characterize, testing and production of various materials and also with the knowledge of new technologies in production and processing of new and advanced engineering materials.

The Department continues to strive for excellence and develop continuously through progress and improvement to establish itself as one of the leading Departments of education and research.

The Faculty Members



Dr. Anup Kumar Keshri
Asst. Professor & Head of
Department
PhD: Florida International
University,
Miami, Florida, USA
Ph: +91-612-3028184
Email-anup@iitp.ac.in,
anup.kumar.keshri@gmail.com



Dr. Anirban Chowdhury
Associate Professor
PhD: University of Leeds,
UK
Ph: +91-612-3028183
Email: anirc@iitp.ac.in
Personal Webpage:-
[http://www.researcher-
id.com/](http://www.researcher-id.com/)



Dr. Dinesh Kumar Kotneer
Asst. professor
PhD: IIT Kharagpur
Ph: +91-612-2552185
Email: dinesh@iitp.ac.in



Dr. Tamoghna Chakrabarti
Asst. Professor
PhD: Indian Institute of
Science Bangalore
Ph: +91-612-302-8745
Email: tamoghna@iitp.ac.in
Webpage: [https://sites.goo-
gle.com/
view/tamoghna](https://sites.google.com/view/tamoghna)



Dr. Devinder Yadav
Asst. Professor
PhD: IIT Madras, 2015
Ph: +91-612-302-8752
Email: devinder@iitp.ac.in



Dr. Ajay Kumar Kalyani
Asst. Professor
PhD: Indian Institute of Science
Metallurgical and Materials
Engineering
Ph: 0612-302-8814
Email: ajay.kalyani@iitp.ac.in

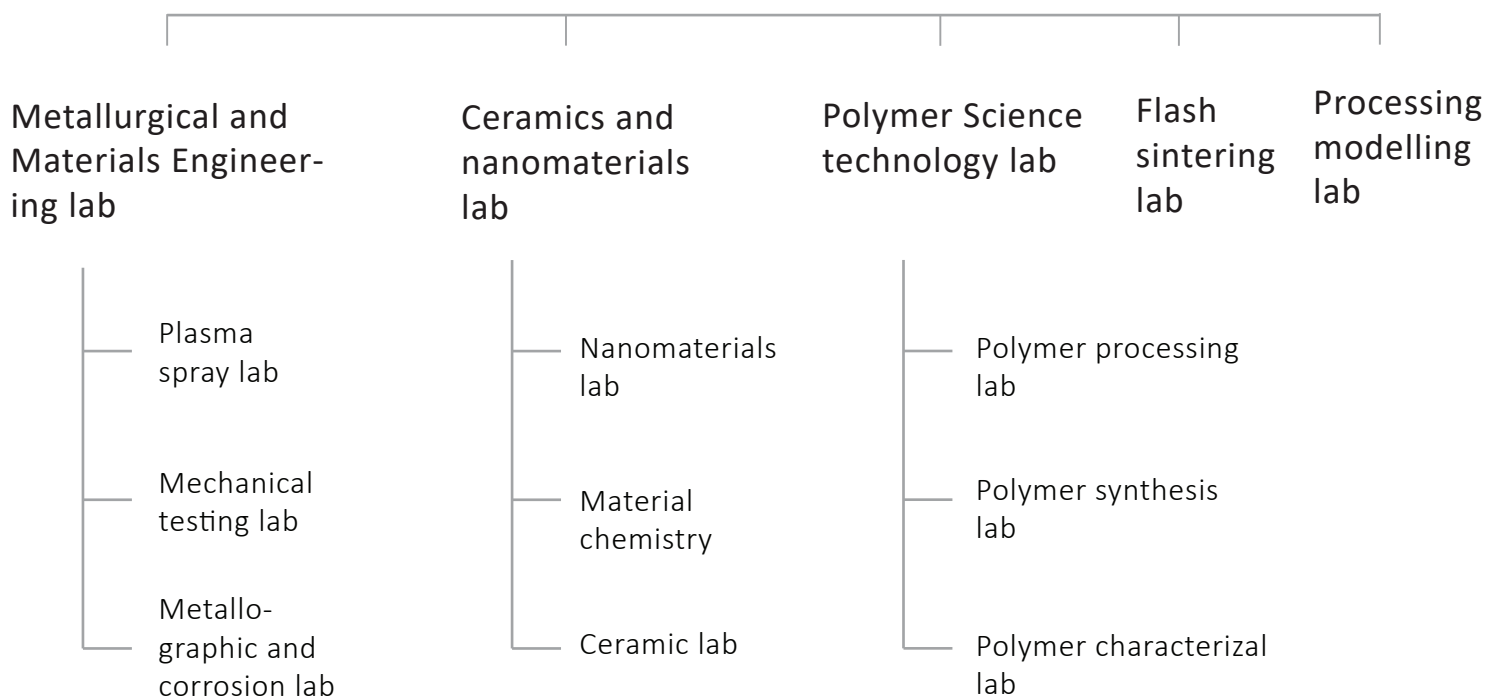
"It is the supreme art of the teacher to awaken joy in creative expression and knowledge."

- Albert Einstein

State of Art Lab Facilities

“MME department in IITP is fully equipment with modern & sophisticated instruments for materials progressing characterisation, which makes students fully learned and gives them hands-on experience for both industries and academics. ”

LABS



Coating Laboratory and Allied Labs



Acoustic chamber- inside view



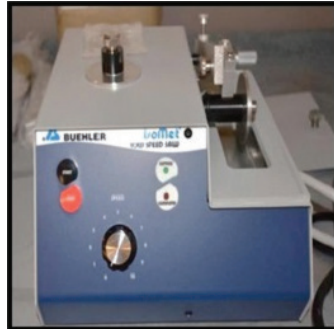
Instrumented hardness and scratch tester



Planetary Ball Mill



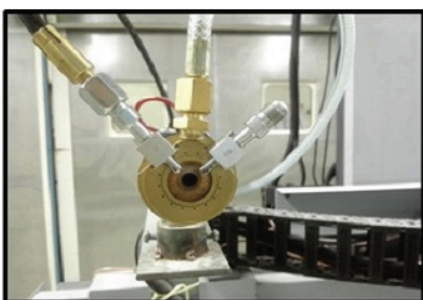
Powder Feeder



Low speed diamond saw



Tribometer



Shrouded Plasma Spraying



Spray drier



High speed diamond saw



Accuraspray



Helium Gas Pycnometer



Hall flow meter

Polymer Science and Allied Labs



Haake Rheomix



Two roll mixing mill



Compression molding machine



Dynamic mechanical analyzer (DMA)



Fourier transform infrared spectroscopy (FTIR)



Simultaneous thermal analyzer (TG-DSC)

Flash Sinthering and Material Modelling Lab



Drum milling machine



Custom built furnace:
950 C



Weighing balance



Ultra Sonicator



Furnace temperature
with 1300 C



Polishing machine

Material Chemistry and Allied labs



Digital ultrasonic cleaner



Milli Q water purification system



Polishing machine



UV spectrometer



Jar mill



Spin coater



Unidirectional hydraulic press



Fume Hood



NETZSCH DSC-TGA



RIGAKU XRD Diffractometer

Our Funders

Our Research scholars are working on various projects which were funded by government as well as private organizations like Tata Steel, ISRO, BHEL, Carborundum Inc., Denka etc.



Points of Pride

- State of the art, **Plasma Spray Coatings Laboratory, PSCL**.
- Department fundings have increased by approx. **70.65%**, which talks about the growth of the department at a rapid rate.
- Department has **1 international** and **11 national** patents.
- Our alumni has excelled both in the field of **placements** and **research**.
- Our students have also bagged off campus placements and are **working at various R&D labs across India**. When it comes to research, our students are pursuing **research in various IIT, IISc, foreign universities like UBC, Cannada, etc.**

Achievements

12 Patents

Approx. fundings of

5.7 Crores

Journals include:

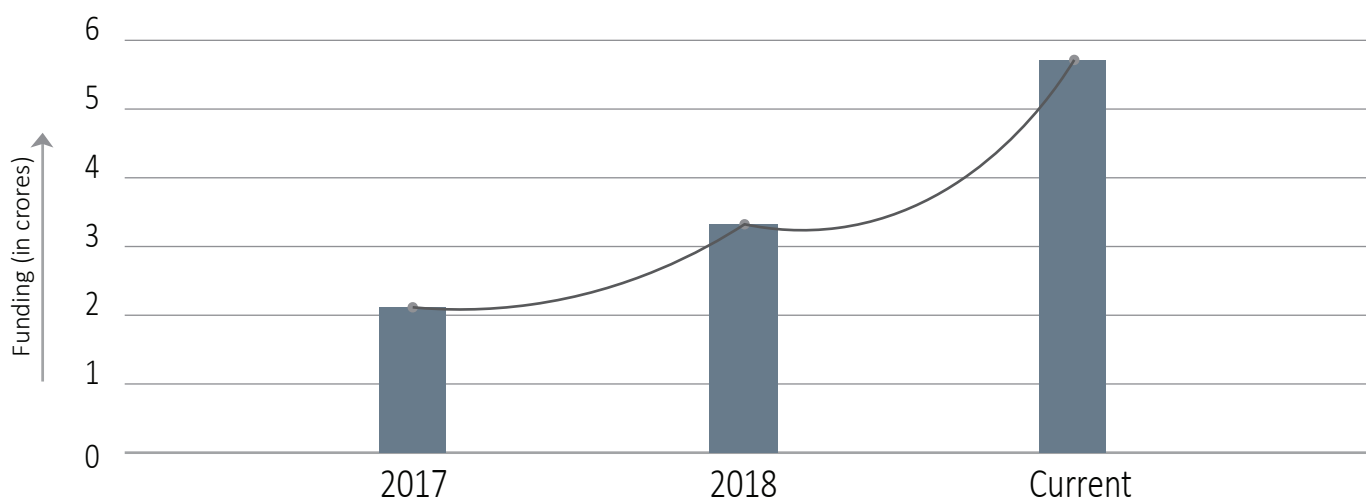
- | | |
|----------------------------------|---|
| - Acta Materialia | - Materials Characterization |
| - Acta Biomaterialia | - Journal of Alloys and Compounds |
| - Composites Part B: Engineering | - Ultrasonics Sonochemistry |
| - Electrochimica Acta | - Applied Surface Science |
| - Scripta Materialia | - Polymer |
| - European Polymer Journal | - Journal of the American Ceramic Society |

Publications

The Department has published over

160 Papers

in many renowned and prestigious journals.



Curriculum

Materials Science and Engineering (MSE) is an interdisciplinary field of science and engineering which investigates the relationship between the structure, property and processing of materials useful for various influence its properties. It is a discipline that enables both the creation and application of materials in society. Materials scientists and engineers develop materials for new applications, improve existing materials to enhance performance and evaluate ways in which different materials can be used together. The M.Tech. degree is designed in a way to provide a holistic view on all the classes of materials including metals, ceramics and polymers. The program is intended to provide in-depth knowledge in the fundamentals, analysis and structure-property correlation of various materials system. The courses will be conducted by faculties from the Department of Materials Science & Engineering (MSE). There will be options also for taking elective courses from within and outside the department. Moreover, specialists from overseas and experts from industries will be invited to lecture for a few classes in selected modules



CORE SUBJECTS

1. Nano-structured material
2. Material processing technology
3. Advance material characterisation techniques
4. Structural and functional properties of material

ELECTIVE SUBJECTS

1. Thermodynamics and phase diagrams
2. Surface engineering
3. Rubber science and technology
4. Alloy development and heat treatment
5. Composite science and technology

LAB COURSES

1. Microstructure and phase analysis lab
2. Materials Characterisation lab

Batch Profile Summary

Student distribution discipline wise PG Specialisation

Polymer Science and Engineering 7.7%

Ceramic Engineering 15.4%

Plastic Engineering 7.7%

Mechanical Engineering 69.2%



AKASH KUMAR
B.E in Mechanical Engg.
Work experience: NIL



BRIJESH SAHOO
B.Tech in Plastic Engg.
Work experience: NIL



NITISH KUMAR
B.Tech
Mechanical Engg.
Work experience: 6 months
[GET (Mechanical), JKTyres Ltd.]



RISHAV RAJ
B.E in Mechanical Engg.
Work experience: 22 months
(Asst. Manager, CEAT Ltd.)



SUBHADIP BHANDARI
B.E in Ceramic Engg.
Work experience: 22 months
(Asst. Manager, CEAT Ltd.)



SHUBHAM KUMAR
B.E in Mechanical Engg.
Work experience: NIL



RAVI RANJAN
B.Tech in Mechanical Engg.
Work experience: NIL



GEORGE VARGHESE P J
B.Tech
Polymer Science and Engg.
Work experience: 22 months
(Asst. Manager, CEAT Ltd.)



PAPPU KUMAR
B.Tech in Mechanical Engg.
Work experience: NIL



RAHUL SINGH
B.Tech in Mechanical Engg.
Work experience: NIL



RAHUL KUMAR
B.Tech in Mechanical Engg.
Work experience: NIL



SAURABH SHARMA
B.E in Ceramic Engg.
Work experience: NIL



SURAJ KUMAR
B.E in Mechanical Engg.
Work experience: NIL



CONTACT US

Training and Placement Cell
Email ID: tpc@iitp.ac.in

Dr. Jose V Parambil
Professor-in-charge
Training and Placement Cell
Email ID: pic_tnp@iitp.ac.in

Dr. Dinesh Kumar Kotnees
Department in-charge
Department of Metallurgical
and Materials Engineering
Email ID: dinesh@iitp.ac.in

Mr. George Varghese P J
Student Co-ordinator (M.Tech)
Training and Placement Cell
Email ID:
george_1911mm04@iitp.ac.in